AQ9.2: Activity Questions 2 - Not Graded

This assignment will not be graded and is only for practice.
Partial Derivatives:

Level 1:

What is the rate of change of $f(x, y, z) = 5x^2 - 6xy + 3z^2 - 2yz$f(x, y, z) = 5x2 − 6xy + 3z2 − 2yz at (1,2,-1) with respect to zz?

No, the answer is incorrect.
Score: 0
Accepted Answers:
(Type: Numeric) -10
1 point
1 point
Let $f: R^2 \rightarrow R$f : R2 → R be defined as $f(x, y) = x^3 + 3xy$f(x, y) = x3 + 3xy. Identify the correct option with respect to the partial derivatives of ff.

- [ ] $\frac{\partial f}{\partial x} = \frac{\partial f}{\partial y}$∂x∂f = ∂y∂f
- [ ] $\frac{\partial f}{\partial x}(0, 0) = \frac{\partial f}{\partial y}(0, 0)$∂x∂f(0, 0) = ∂y∂f(0, 0)
- [ ] $\frac{\partial f}{\partial x}(1, 1) = \frac{\partial f}{\partial y}(1, 1)$∂x∂f(1, 1) = ∂y∂f(1, 1)
- [ ] $\frac{\partial f}{\partial x}(1, 0) = \frac{\partial f}{\partial y}(1, 0)$∂x∂f(1, 0) = ∂y∂f(1, 0)
- [ ] $\frac{\partial f}{\partial x} = \frac{\partial f}{\partial y} + 3$∂x∂f = ∂y∂f + 3

No, the answer is incorrect.
Score: 0
Accepted Answers:
$\frac{\partial f}{\partial x}(0, 0) = \frac{\partial f}{\partial y}(0, 0)$∂x∂f(0, 0) = ∂y∂f(0, 0)
$\frac{\partial f}{\partial x}(1, 0) = \frac{\partial f}{\partial y}(1, 0)$∂x∂f(1, 0) = ∂y∂f(1, 0)
1 point
Let $f(x, y) = \frac{xy}{x^2 + y}$f(x, y) = x2+yxy. Identify the correct options with respect to the partial derivatives of ff.

- [ ] $\frac{\partial f}{\partial x}$∂x∂f and $\frac{\partial f}{\partial y}$∂y∂f are not defined at $(0, 0)$(0, 0)
- [ ] $\frac{\partial f}{\partial x}(1, 1) = 0, \frac{\partial f}{\partial y}(1, 1) = 1/4$∂x∂f(1, 1) = 0, ∂y∂f(1, 1) = 1/4
- [ ] $\frac{\partial f}{\partial x}(1, 1) = 0, \frac{\partial f}{\partial y}(1, 1) = 0$∂x∂f(1, 1) = 0, ∂y∂f(1, 1) = 0
- [ ] $\frac{\partial f}{\partial x}(1, 0) = 0, \frac{\partial f}{\partial y}(1, 0) = 1$∂x∂f(1, 0) = 0, ∂y∂f(1, 0) = 1

No, the answer is incorrect.
Score: 0
Accepted Answers:

$\frac{\partial f}{\partial x}$∂x∂f and $\frac{\partial f}{\partial y}$∂y∂f are not defined at $(0, 0)$(0, 0)
$\frac{\partial f}{\partial x}(1, 1) = 0, \frac{\partial f}{\partial y}(1, 1) = 1/4$∂x∂f(1, 1) = 0, ∂y∂f(1, 1) = 1/4
$\frac{\partial f}{\partial x}(1, 0) = 0, \frac{\partial f}{\partial y}(1, 0) = 1$∂x∂f(1, 0) = 0, ∂y∂f(1, 0) = 1
Let $f: R^2 \rightarrow R$f : R2 → R be defined as $f(x, y) = x^2 + 4xy - y^2$ f(x, y) = x2 + 4xy − y2. Suppose $\frac{\partial f}{\partial x} = \frac{\partial f}{\partial y}$∂x∂f = ∂y∂f at $(12, k)$(12, k). Find the value of kk

No, the answer is incorrect.
Score: 0
Accepted Answers:
(Type: Numeric) 4
1 point

Level 2:

1 point
Suppose $f(x, y, z) = xy + yz + zx - 3xyz$f(x, y, z) = xy + yz + zx − 3xyz is a multivariable function defined on domain $D \subset R^3$ D ⊂ R3. Which of the following is(are) correct?

☐ ff is a scalar valued multivariable function.

☐ The rate of change of the function ff at (0,1,1) with respect to xx is 1.

☐ The rate of change of the function ff at (2,2,2) with respect to xx is same as the rate of change of the function ff at (2,2,2) with respect to zz.

☐ The rate of change of the function ff at (1,3,5) with respect to yy is -9.

No, the answer is incorrect.
Score: 0
Accepted Answers:
ff is a scalar valued multivariable function.
The rate of change of the function ff at (2,2,2) with respect to xx is same as the rate of change of the function ff at (2,2,2) with respect to zz.
The rate of change of the function ff at (1,3,5) with respect to yy is -9.
1 point
Suppose $f: D \longrightarrow R$f : D ⟶ R is a multivariable function defined on domain $D \subset R^2$ D ⊂ R2 and also satisfying the following conditions:

(i) $\frac{\partial f}{\partial x} = 6x + 4xy + 3y$∂x∂f = 6x + 4xy + 3y
(ii) $\frac{\partial f}{\partial y} = 2x^2 + 3x - 3y$∂y∂f = 2x2 + 3x − 3y
Which of the following functions can not be ff?

☐ $f(x, y) = 3x^2 + 2x^2 y + 3xy - \frac{3}{2}y^2$ f(x, y) = 3x2 + 2x2y + 3xy − 23y2

☐ $f(x, y) = 3x^2 + 2x^2 y - 3xy - \frac{3}{2}y^2$ f(x, y) = 3x2 + 2x2y − 3xy − 23y2

☐ $f(x, y) = 3x^2 - 3xy - 2x^2 y + 3y^2$ f(x, y) = 3x2 − 3xy − 2x2y + 3y2

☐ $f(x, y) = 3x^2 + 3y^2 + 3xy$f(x, y) = 3x2 + 3y2 + 3xy

No, the answer is incorrect.
Score: 0
Accepted Answers:
$f(x, y) = 3x^2 + 2x^2 y - 3xy - \frac{3}{2}y^2$ f(x, y) = 3x2 + 2x2y − 3xy − 23y2
$f(x, y) = 3x^2 - 3xy - 2x^2 y + 3y^2$ f(x, y) = 3x2 − 3xy − 2x2y + 3y2
$f(x, y) = 3x^2 + 3y^2 + 3xy$ f(x, y) = 3x2 + 3y2 + 3xy

Let $f: R^2 \rightarrow R^2$ f : R2 → R2 be given by $f(x, y) = (P(x, y), Q(x, y))$ f(x, y) = (P(x, y), Q(x, y)), where $P(x, y) = 3x^2 y, Q(x, y) = 5x + y^3$ P(x, y) = 3x2y, Q(x, y) = 5x + y3. Consider a matrix $A = \begin{bmatrix} \frac{\partial P}{\partial x} & \frac{\partial P}{\partial y} \\ \frac{\partial Q}{\partial x} & \frac{\partial Q}{\partial y} \end{bmatrix}$ A = [∂x∂P ∂x∂Q∂y∂P ∂y∂Q]. If the determinant of A A is $axy^3 - bx^2$ axy3 − bx2 where $a, b \in R$ a, b ∈ R, then find the value of $a - b$ a − b.

No, the answer is incorrect.
Score: 0
Accepted Answers:
(Type: Numeric) 3
1 point

Let $f: R^2 \rightarrow R^2$ f : R2 → R2 be given by $f(x, y) = (P(x, y), Q(x, y))$ f(x, y) = (P(x, y), Q(x, y)), where $P(x, y) = 6x + y^2, Q(x, y) = 3x^2 + 2y$ P(x, y) = 6x + y2, Q(x, y) = 3x2 + 2y. Consider a matrix $A = \begin{bmatrix} \frac{\partial P}{\partial x} & \frac{\partial P}{\partial y} \\ \frac{\partial Q}{\partial x} & \frac{\partial Q}{\partial y} \end{bmatrix}$ A = [∂x∂P ∂x∂Q∂y∂P ∂y∂Q]. If $det(A) = 0$ det(A) = 0, then find the value of xy

No, the answer is incorrect.
Score: 0
Accepted Answers:
(Type: Numeric) 1
1 point

Let $f: R^2 \rightarrow R$ f : R2 → R be defined as $f(x, y) = e^{4(x+y)} sin(7x - 4y)$
f(x, y) = e4(x+y)sin(7x − 4y). If $4\frac{\partial f}{\partial x} + 7\frac{\partial f}{\partial y} = k.f(x, y)$ 4∂x∂f + 7∂y∂f = k.f(x, y) where $k \in R$ k ∈ R, then find the value of k k.

No, the answer is incorrect.
Score: 0
Accepted Answers:
(Type: Numeric) 44
1 point

Check Answers
Your score is: 0/9